

Full solution packet for arXiv:2306.16026

The critical four-corner Cantor dust is common hypercyclic

Status. full_solution_likely_valid. This packet gives a positive answer to Problem 6.1 of Fernando Costa Jr., *Self-similar fractals and common hypercyclicity*, arXiv:2306.16026.

Source problem. Let $\mathcal{C} \subset \mathbb{R}_+^2$ be a Cantor dust with uniform dissection ratio $1/4$. Does

$$(e^\lambda B \times e^\mu B)_{(\lambda, \mu) \in \mathcal{C}}$$

acting on $c_0 \times c_0$ have a common hypercyclic vector? The problem appears as Problem 6.1 on page 36 of the current arXiv PDF.

fractals, like the Cantor dust for example. A “break of nature” happens with these fractals: we go from having no jumps to having jumps on every single step. The simplest version of this problem can be stated as follows.

Problem 6.1. Let $\mathcal{C} \subset \mathbb{R}_+^2$ be a Cantor dust with uniform dissection ratio $1/4$. Does the family of products of Rolewicz operators $(e^\lambda B \times e^\mu B)_{(\lambda, \mu) \in \mathcal{C}}$ acting on $c_0 \times c_0$ have a common hypercyclic vector?

Notice that [4, Example 4.9] in combination with Borichev’s example give the answer for all Cantor dusts with uniform dissection ratio $c \neq \frac{1}{4}$. The limit case $c = \frac{1}{4}$ remains open.

When it comes to self-similar connected fractals, one can notice that the mere existence of a homogeneously ordered family of partitions provides a family of curves converging pointwise to the fractal itself. In fact, if $((\Lambda_i)_{i \in I_r^m})_{m \in \mathbb{N}}$ is a homogeneously ordered family of coverings for Λ , then for each $m \in \mathbb{N}$, we fix one point p_i inside each part Λ_i , $i \in I_r^m$, and we link them all following the lexicographical ordering of I_r^m . This defines a polygonal path $P_m = [p_{1, \dots, 1}, \dots, p_{r, \dots, r}]$, for all $m \in \mathbb{N}$. Then $\Lambda = \lim_{m \rightarrow +\infty} P_m$ where the convergence is pointwise. However, although homogeneity and ordering of the family might imply the continuity of the limit curve, calculating its Hölder exponent is a more delicate task. It is, then, much more natural to apply our result by taking into account the number of similarities and their contraction ratio rather than looking at it as a Hölder curve.

Particularly for self-similar fractals that are not uniformly contracting, we

Definitions and the characterization used

Here a uniform four-corner Cantor dust means the usual attractor obtained from an axis-parallel square $Q_\emptyset \subset (0, \infty)^2$ by retaining, inside every square, its four corner subsquares of side one quarter of the parent side. Write $L > 0$ for the side of $Q_\emptyset$. At level m , the dust is contained in 4^m axis-parallel squares Q_w , $w \in \{0, 1, 2, 3\}^m$, each of side $L4^{-m}$.

We use Theorem 2.2 of Bayart–Costa–Menet [2]. In the case $d = 2$, it says that the family

$$(e^{\lambda(1)} B \times e^{\lambda(2)} B)_{\lambda \in \Lambda}$$

on $c_0 \times c_0$ has a common hypercyclic vector if and only if the following condition holds. For every $\tau > 0$, every $N \geq 1$, and every compact $K \subset \Lambda$, there are

$$N \leq n_1 < n_1 + N \leq n_2 < \dots < n_{q-1} + N \leq n_q$$

and tags $\lambda_k \in (0, \infty)^2$ such that

$$K \subset \bigcup_{k=1}^q \prod_{i=1}^2 \left[\lambda_k(i) - \frac{\tau}{n_k}, \lambda_k(i) \right], \quad (1)$$

$$\lambda_{k+1}(i)n_{k+1} - \lambda_k(i)n_k \geq N \quad (i = 1, 2; 1 \leq k < q). \quad (2)$$

The tags need not themselves belong to Λ .

Proof intuition

A level- m listing of all 4^m dust squares has a logarithmic accumulation of jumps: each generation contributes a bounded amount of variation. This is exactly why using one fixed resolution loses at the endpoint.

The remedy is to use many short blocks at gradually increasing resolutions. At resolution m , cover only a symbolic portion comparable to τ/m . The accumulated jump variation within such a block is then uniformly bounded, so the times required by (2) grow by at most a fixed factor. Moving from m to $m + s$, for one fixed integer s , supplies exactly that fixed-factor room. The portions covered are comparable to

$$\frac{\tau}{m_0}, \quad \frac{\tau}{m_0 + s}, \quad \frac{\tau}{m_0 + 2s}, \quad \dots$$

Their sum diverges. Consequently, after finitely many blocks the whole symbolic Cantor dust is covered. This converts the unavoidable jumps into a harmonic cost instead of trying to eliminate them.

Two estimates

Order the words in $\{0, 1, 2, 3\}^m$ lexicographically. This order is consistent under refinement: every level- ℓ cylinder is a consecutive block of $4^{m-\ell}$ level- m cylinders. Let u_w be the upper-right corner of Q_w .

Lemma 1 (Variation of a consecutive dust block). *Let $w_r, \dots, w_{r+q-1}$ be q consecutive level- m words. Then*

$$\sum_{j=r}^{r+q-2} \|u_{w_{j+1}} - u_{w_j}\|_{\infty} \leq 4L m \frac{q}{4^m} + \frac{8L}{3}.$$

If one prepends one arbitrary tag from $Q_{\emptyset}$, the right-hand side may be replaced by

$$4L m \frac{q}{4^m} + \frac{11L}{3}.$$

Proof. For two consecutive level- m words, let h be the length of their common prefix. Their tags lie in the same level- h square, and hence their distance is at most $L4^{-h}$. Charge this jump to every level $\ell = h + 1, \dots, m$, with charge $L4^{-(\ell-1)}$.

For a fixed ℓ , a consecutive block of q level- m words crosses at most

$$\frac{q}{4^{m-\ell}} + 2$$

boundaries of level- ℓ blocks. Summing the charges gives

$$\begin{aligned} \sum_{j=r}^{r+q-2} \|u_{w_{j+1}} - u_{w_j}\|_\infty &\leq \sum_{\ell=1}^m L 4^{-(\ell-1)} \left(\frac{q}{4^{m-\ell}} + 2 \right) \\ &\leq 4L m \frac{q}{4^m} + \frac{8L}{3}. \end{aligned}$$

All tags lie in the initial square, so prepending one tag adds at most L in the sup norm. $\square$

Lemma 2 (Scheduling tags). *Suppose all coordinates of the tags $\lambda_0, \lambda_1, \dots, \lambda_q$ are at least $a > 0$. Starting from an integer n_0 , define n_j to be the least integer satisfying*

$$n_j \geq n_{j-1} + N, \quad n_j \lambda_j(i) \geq n_{j-1} \lambda_{j-1}(i) + N \quad (i = 1, 2).$$

There is a constant $A = A(a, N) \geq 1$ such that

$$n_q \leq (n_0 + Aq) \exp \left(A \sum_{j=1}^q \|\lambda_j - \lambda_{j-1}\|_\infty \right).$$

Proof. Put $d_j = \|\lambda_j - \lambda_{j-1}\|_\infty$. The definition and the lower coordinate bound give, for a sufficiently large $A = A(a, N)$,

$$n_j \leq n_{j-1} + A(1 + n_{j-1} d_j) = (1 + A d_j) n_{j-1} + A.$$

Iterating this affine inequality and using $\prod_j (1 + A d_j) \leq \exp(A \sum_j d_j)$ yields the claim. $\square$

Main result

Theorem 3. *Let $\mathcal{C} \subset \mathbb{R}_+^2$ be a uniform axis-parallel four-corner Cantor dust with dissection ratio $1/4$. Then*

$$\bigcap_{(\lambda, \mu) \in \mathcal{C}} \text{HC}(e^\lambda B \times e^\mu B) \neq \emptyset$$

on $c_0 \times c_0$.

Proof. Fix $\tau > 0$ and $N \geq 1$. It is enough to verify (1)–(2) for all of $\mathcal{C}$, rather than for one compact subset. Replacing τ by a smaller number only strengthens the required cover, so put

$$\varepsilon = \min\{\tau, L\}, \quad \theta = \frac{\varepsilon}{L} \in (0, 1].$$

All coordinates of all level tags are bounded below by some $a > 0$. Let $A = A(a, N)$ be supplied by Lemma 2, and set

$$V_* = \frac{23L}{3}, \quad P = e^{AV_*}, \quad \eta = \frac{1}{4P}.$$

Choose an integer $s \geq 1$ so that

$$4^s \geq 2P.$$

Choose m_0 sufficiently large that

$$N \leq \eta \theta 4^{m_0}, \quad m_0 \geq 4PA, \quad \frac{\theta 4^{m_0}}{m_0} \geq 2. \quad (3)$$

We construct finitely many consecutive symbolic blocks. At stage j , use resolution

$$m_j = m_0 + js.$$

The endpoint of every earlier stage is a level- m_j grid point. If R_j level- m_j cylinders remain, take the next

$$q_j = \min \left\{ R_j, \left\lfloor \frac{\theta 4^{m_j}}{m_j} \right\rfloor \right\} \quad (4)$$

cylinders in lexicographic order, and tag them by their upper-right corners.

We schedule the tags consecutively using Lemma 2. For the very first tag, use a dummy preceding tag equal to it and dummy time N ; the resulting first actual time is at least $2N$, which is harmless. At every later stage, the preceding tag and time are the last ones from the previous stage.

We claim inductively that, at the start of a stage of resolution m , its incoming time n_{in} satisfies

$$n_{\text{in}} \leq \eta \theta 4^m. \quad (5)$$

This holds initially by (3). From (4) and $\theta \leq 1$,

$$m \frac{q_j}{4^m} \leq \theta \leq 1.$$

Lemma 1, including the incoming transition, therefore bounds the tag variation during this stage by

$$4L + \frac{11L}{3} = V_*.$$

Lemma 2 and $m \geq m_0 \geq 4PA$ give

$$n_{\text{out}} \leq P(n_{\text{in}} + Aq_j) \quad (6)$$

$$\leq P\eta\theta 4^m + PA \frac{\theta 4^m}{m} \leq \frac{1}{2}\theta 4^m. \quad (7)$$

In particular, every time used during this stage satisfies

$$n_k \leq \theta 4^m.$$

Since its cylinder square has side $L4^{-m}$, we have

$$L4^{-m} \leq \frac{\varepsilon}{n_k} \leq \frac{\tau}{n_k}.$$

Thus the rectangle in (1), with tag equal to the upper-right corner, contains the whole part of $\mathcal{C}$ in that cylinder.

Moreover, (7) and $4^s \geq 2P$ imply

$$n_{\text{out}} \leq \frac{1}{2}\theta 4^m \leq \eta\theta 4^{m+s},$$

which proves the invariant for the next stage. By construction of the times, their gaps are at least N and (2) holds.

It remains to prove that only finitely many stages are needed. If a stage does not finish the cover, then (3) and (4) imply

$$\frac{q_j}{4^{m_j}} \geq \frac{\theta}{2m_j}.$$

But

$$\sum_{j=0}^{\infty} \frac{\theta}{2(m_0 + js)} = \infty.$$

Hence the successive symbolic portions cannot all fit inside a total symbolic interval of length 1. Some finite stage must take all remaining cylinders. The resulting finite concatenation covers $\mathcal{C}$ and satisfies (1)–(2). Theorem 2.2 of Bayart–Costa–Menet now gives the required common hypercyclic vector. $\square$

Corollary 4. *The same conclusion holds with c_0 replaced by ℓ_p for any $1 \leq p < \infty$.*

Proof. The characterization used above holds verbatim for ℓ_p . $\square$

Verification notes

- The proof uses only the exact necessary-and-sufficient characterization in Theorem 2.2 of [2]; no unproved dynamical criterion is introduced.
- The critical accounting is explicit: a block of symbolic length $q/4^m$ costs at most $O(1 + mq/4^m)$ tag variation. Choosing $q/4^m \asymp \theta/m$ makes this cost uniform.
- The stage-time invariant is checked in (5)–(7). The fixed resolution increment s absorbs the uniform multiplicative scheduling cost, and the harmonic divergence proves finite termination.
- Exploratory code in `code/finite_traversal_probe.cpp` implements a different greedy Hilbert-prefix schedule. For $\tau = 0.1$, it advanced through 0.877709756 of symbolic mass after 10^9 pieces. This was a stress test that motivated the block proof; it is not used as evidence for any formal step.

Novelty check and limitations

On 23 July 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2306.16026 and for combinations of `Cantor dust`, `dissection ratio 1/4`, `common hypercyclicity`, and `Rolewicz`. The local arXiv source corpus was also searched. Exact-phrase web searches restricted to arXiv returned the source paper but no later paper claiming to answer Problem 6.1. This is a bounded novelty check, not a guarantee of priority.

The theorem treats the standard uniform axis-parallel four-corner dust (and its positive axis-parallel translates and scalings), which is the Cantor-dust construction in the source problem. It does not claim the same result for arbitrary rotated or non-corner self-similar sets, or for the source paper’s other open problems.

Human-review recommendation. Review as a claimed full positive solution of Problem 6.1. The main points to check are Lemma 1’s multiscale boundary count and the stage invariant in (7). Neither step depends on the exploratory computation.

References

- [1] F. Costa Jr., *Self-similar fractals and common hypercyclicity*, arXiv:2306.16026, Problem 6.1.
- [2] F. Bayart, F. Costa Jr., and Q. Menet, *Common hypercyclic vectors and dimension of the parameter set*, Indiana Univ. Math. J. **71** (2022), 1763–1795; arXiv:2103.13152, Theorem 2.2.